



Pixels to Viewport Coordinates

? The Y axis is flipped because in screen coordinates "y = 0" is the top. In the 3D world, the "y = 0" is the bottom, so the Y axis is flipped to prevent the image from being upside down.

$$y_ratio = 1.0 - (2.0 * (y + 0.5) / WIN_HEIGHT);$$

Screen (5x5 pixels):

●	●	●	●	●
●	●	●	●	●
●	●	●	●	●
●	●	●	●	●
●	●	●	●	●

x=0 → viewport_x = -0.8

x=1 → viewport_x = -0.4

x=2 → viewport_x = 0.0

x=3 → viewport_x = 0.4

x=4 → viewport_x = 0.8

Each ● is where the ray goes through.

Step 1: Start with pixel coordinates

```
Screen (pixels):  
(0,0) → (WIN_WIDTH-1, WIN_HEIGHT-1)
```

Step 2: Add 0.5 to get pixel center

We shoot rays through the center of each pixel, not the corner:

```
pixel_center_x = x + 0.5  
pixel_center_y = y + 0.5
```

Step 3: Normalize to [0,1] range

Divide by total width and height:

```
normalized_x = (x + 0.5) / WIN_WIDTH    // Range [0, 1]  
normalized_y = (y + 0.5) / WIN_HEIGHT   // Range [0, 1]
```

Step 4: Scale to [-1,1] range

Multiply by 2 and subtract 1:

```
viewport_x = 2 * normalized_x - 1    // Maps [0,1] → [-1,1]  
viewport_y = 2 * normalized_y - 1    // Maps [0,1] → [-1,1]
```